1. Data Architecture

To support data-driven decision making and machine learning applications, a structured data architecture is required that consolidates operational data from multiple sources and prepares it for analysis. The proposed architecture organizes data collection, storage, and transformation into a pipeline that enables efficient analytics and model development.

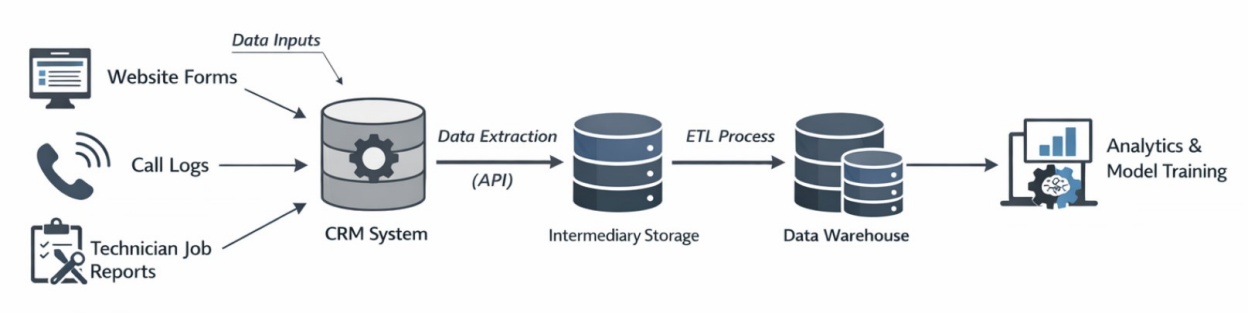

Figure 1: Data Engineering Pipeline for Analytics and Machine Learning

The overall pipeline consists of five main stages: data sources, CRM integration, data extraction, intermediary storage, and a data warehouse for analytics and machine learning.

The system receives data from several operational sources, including website forms, call logs, and technician job reports. These data sources capture information about customer inquiries, sales interactions, and service operations. All incoming data is stored in the CRM system, which acts as the central platform for managing leads, customer records, and job information.

To enable analytics and machine learning, relevant data is periodically extracted from the CRM using API-based data extraction. The extracted data is first stored in an intermediary storage layer, which serves as a staging area for processing. At this stage, basic transformations such as data cleaning, formatting, and merging of related datasets are performed.

After transformation, the processed data is loaded into a data warehouse, where it is organized into structured tables optimized for analytical queries. This environment supports business intelligence reporting and provides the dataset required for analytics and model training.

By organizing operational data through this pipeline, the company creates a scalable data infrastructure that enables both analytics and AI-driven insights.

2. Machine Learning Use Case: Lead Conversion Prediction

A machine learning model can be used to predict the probability that a lead will convert into a paying customer. This prediction helps the sales team prioritize high-value leads, respond faster, automate some tasks based on lead score and reduce missed revenue opportunities.

Features Used

The model will be trained using features derived from historical lead data stored in the CRM. Key features include lead source, which identifies the marketing channel through which the lead was acquired (e.g., website, advertisement, or referral), and service type requested, which indicates the type of service the customer needs.

Several features can be derived from the raw data. Examples include response time to the first contact, total number of follow-up attempts, call count, total_call_duration, lead acquisition channel, time of inquiry, and historical conversion rate of each marketing channel. Interaction-based features such as call frequency or delay between contact attempts can also help capture customer engagement patterns.

Model Choice

For this problem, a tree-based classification model such as XGBoost is selected to predict lead conversion probability. The dataset in this case is inherently sparse, because leads originate from multiple sources (e.g., website forms, call logs, or referrals), meaning many records will have NULL or missing values for certain features. For example, a lead created from a website form may initially have no call-related data. XGBoost performs well in such scenarios because it can naturally handle missing values during tree construction and learn optimal split directions for them. This makes it particularly effective for classification tasks with sparse, tabular business data, while also capturing complex relationships between features that influence lead conversion.

Performance Evaluation

Model performance can be evaluated using metrics such as ROC-AUC, precision, recall, and F1-score. But, main importance will be given to recall. These metrics help ensure that the model accurately identifies high-probability leads while minimizing false negatives. Cross-validation can also be used to verify that the model generalizes well to new data.

3. System Architecture and Automation Workflow

The proposed Lead Intelligence system introduces a data-driven lead management framework that helps the company identify high-value leads, improve follow-up efficiency with automation, and gain visibility into marketing performance. Currently, leads are often contacted late, follow-ups are inconsistent, and management lacks clear insight into which marketing channels generate the best customers. By integrating machine learning with the existing CRM platform, the system can automatically prioritize leads, monitor sales performance, and generate real-time analytics. This approach enables the company to focus sales efforts on the most promising prospects while providing management with actionable insights to improve marketing and operational decisions.

The system architecture is shown below:

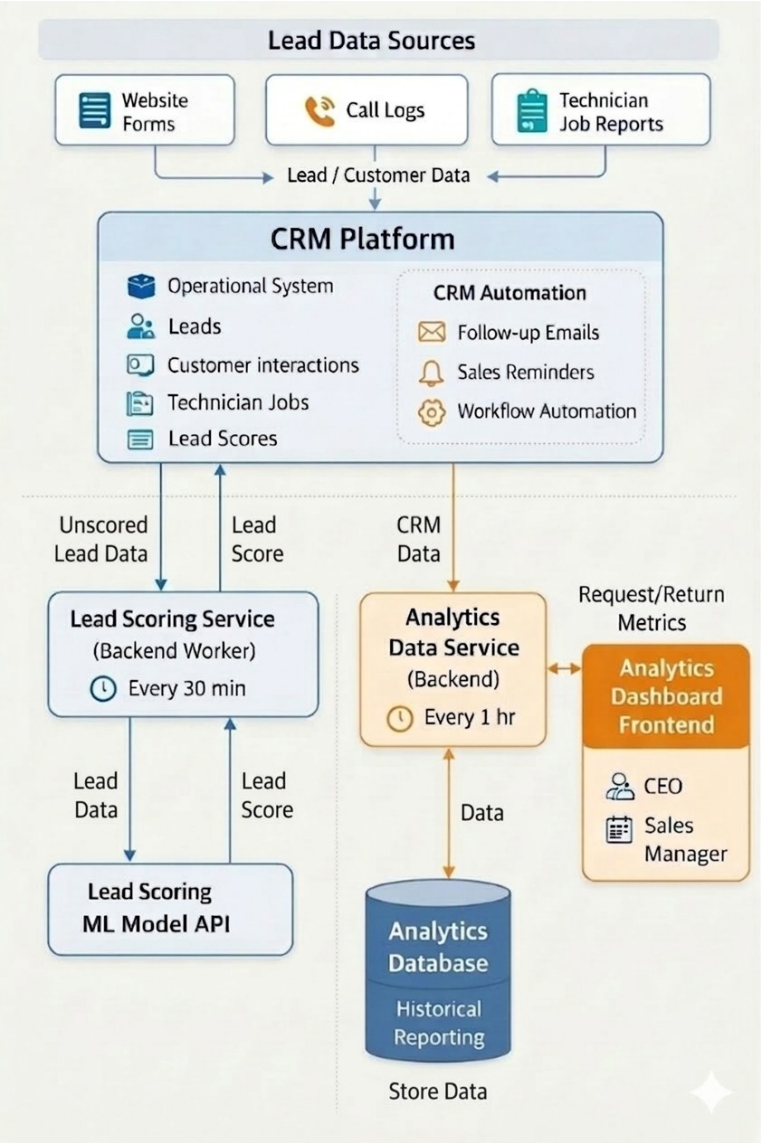

Figure 2: System Architecture

3.1. Unified Data Collection through CRM

Lead data originates from multiple sources such as website forms, call logs, and technician job reports. These data sources feed information into a centralized CRM platform, which acts as the single operational system for storing and managing all lead and customer interactions. The CRM maintains structured records including lead details, communication history, and technician activities. Since the CRM exposes its data through APIs, other system components can access the required information directly from this central source. This ensures that all downstream services operate on consistent and unified lead data.

3.2. Automated Lead Scoring System

To identify high-potential customers, a Lead Scoring Service (backend worker) periodically retrieves leads from the CRM that have not yet been scored. The service sends the relevant lead data to a deployed machine learning model through a Lead Scoring ML Model API. The model predicts the probability that a lead will convert into a paying customer and returns the lead score to the backend service. The backend then updates the CRM record through the CRM API by attaching the predicted lead score. As a result, the CRM interface can automatically highlight high-probability leads, enabling the sales team to prioritize the most promising opportunities.

3.3. Analytics and Operational Intelligence

In addition to real-time lead scoring, the system provides analytics capabilities for management. An Analytics Data Service (backend) periodically retrieves historical lead and interaction data from the CRM and stores it in a dedicated Analytics Database. When users access the analytics dashboard, the frontend requests metrics from the backend service. The backend then queries the analytics database, calculates performance indicators such as conversion rates, channel performance, and sales activity metrics, and returns the results to the dashboard. This allows executives and sales managers to monitor overall business performance through visual reports.

3.4. Automation of Sales Follow-Ups

Automation is implemented directly within the CRM system using the predicted lead scores. When a lead receives a high score, the CRM can automatically trigger follow-up emails or reminders for the sales team to contact the lead promptly. If a salesperson forgets to follow up within a specified time, the CRM can generate automated reminders. This ensures that high-value leads are not missed and that engagement with potential customers remains timely and consistent.

Together, these components create an integrated intelligent system that combines machine learning, analytics, and CRM automation. The result is a streamlined workflow where leads are automatically prioritized, sales teams receive timely reminders, and management gains clear insights into marketing effectiveness and business performance.

4. Business Intelligence Dashboard

The business intelligence dashboard provides management with a clear, data-driven overview of operational and marketing performance. The dashboard displays key metrics such as lead conversion rate, average response time to new leads, and revenue generated per technician, allowing the owner to monitor how effectively leads are being handled and converted into customers. It also includes customer acquisition channel performance, which shows which marketing channels generate the highest number of leads and the most valuable customers.

Additional visualizations include the lead management funnel, highlighting how leads move through stages such as contacted, appointment scheduled, and closed deals, as well as the distribution of lead status. The dashboard also presents marketing channel efficiency, comparing cost per lead with revenue per customer. By consolidating these insights into one interface, the dashboard enables the owner and sales managers to quickly identify operational bottlenecks, evaluate marketing effectiveness, and make informed strategic decisions.

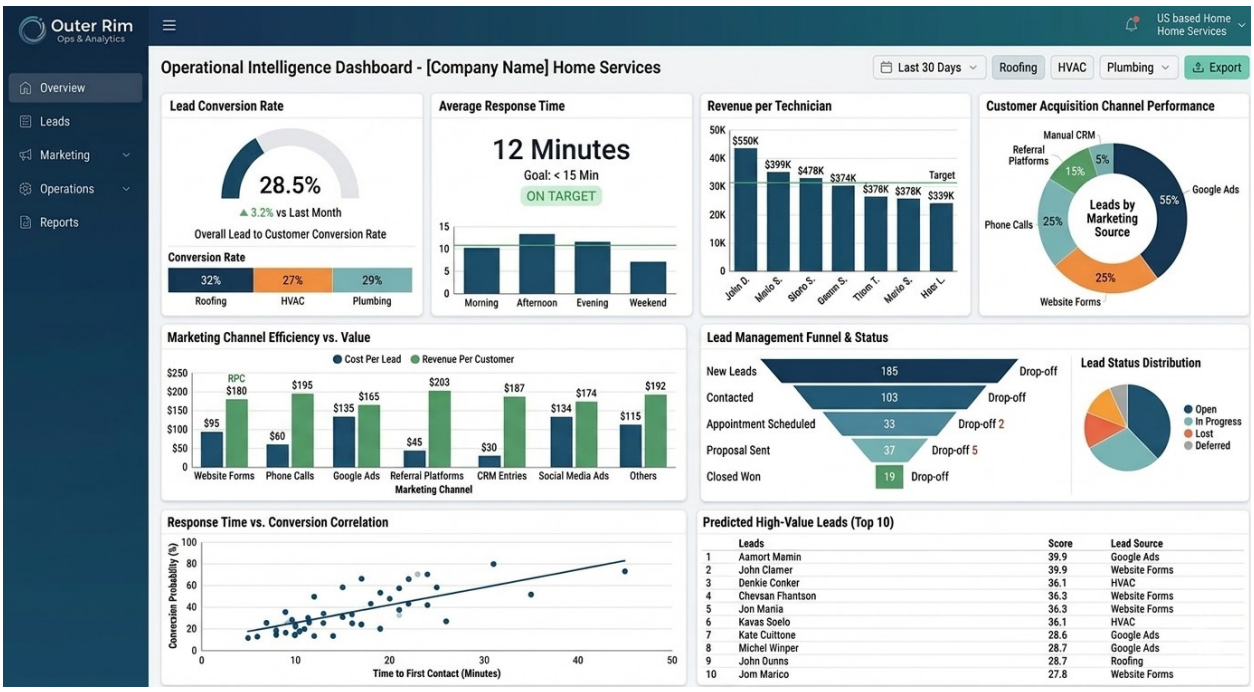

Figure 3: Business Intelligence Dashboard